

Harmain Ali

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Professional Summary

Accomplished Senior Backend Software Engineer with 9+ years of experience designing, building, and operating distributed systems at scale. Deep expertise in backend architecture, data ingestion pipelines, query optimization, and storage solutions for observability and product analytics platforms. Proficient in Go, Java, and SQL with proven track record of leading technical initiatives, mentoring engineering teams, and delivering high-quality software products. Experienced in processing clickstreams, user behavioral data, and multi-source data correlation. Strong technical leader comfortable influencing architecture decisions and remaining a hands-on contributor. Passionate about leveraging AI tools to enhance code quality, problem-solving, and engineering best practices.

Skills

Core Skills:	Go, Java, Python, C++, SQL, PostgreSQL, MySQL, Cassandra, Apache Kafka, Data Ingestion, Query Optimization, Storage Solutions, Distributed Databases, Time-Series Databases, ClickHouse, Elasticsearch, Observability, Product Analytics, Clickstreams, Event Processing, Metrics Collection, Distributed Tracing, Log Aggregation, Real-Time Analytics, Frontend, JavaScript, React, TypeScript, RESTful APIs, GraphQL, Web Development, AWS, Google Cloud Platform, Docker, Kubernetes, Microservices, CI/CD, Infrastructure as Code, Technical Leadership, Mentorship, Architecture Design, Cross-Functional Collaboration, Code Review, Design Reviews, Team Building, AI Coding Tools, Machine Learning, TensorFlow, PyTorch, Prompt Engineering, AI-Enhanced Development
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Professional Experience

Datadog | San Francisco, CA Mar 2023 - Present

Senior Backend Software Engineer

Technologies: Go, Java, SQL, PostgreSQL, Kafka, Kubernetes, AWS, GraphQL

- Architected and deployed core backend systems for observability platform processing 2+ trillion events daily, designing distributed data ingestion pipelines in Go that handle clickstreams and multi-source data correlation with 99.99% reliability. Implemented query optimization strategies reducing average query latency by 40% through advanced SQL indexing and storage solutions leveraging ClickHouse and Cassandra. Led technical architecture decisions for product analytics backend, mentoring team of 6 engineers on distributed systems design patterns and establishing code review standards that improved deployment quality by 35%. Collaborated with frontend teams to design GraphQL APIs exposing analytics capabilities, enabling product managers to build data-driven features while maintaining backend performance SLAs.
- Drove adoption of AI coding tools across engineering team, establishing best practices for prompt engineering and AI-generated code validation that increased development velocity by 25%. Engineered real-time clickstream processing system in Go handling 500K events per second, implementing fault-tolerant data ingestion with automatic retry logic and dead-letter queue management. Optimized storage solutions by migrating time-series data to specialized databases, reducing storage costs by 30% while improving query performance for product analytics use cases. Mentored junior engineers on observability best practices, including distributed tracing instrumentation and metrics collection strategies.
- Designed and implemented multi-tenant data isolation architecture supporting 1000+ customer accounts, ensuring data security and query performance through intelligent partitioning and storage solutions. Established observability standards for backend services using Datadog monitoring, creating dashboards and alerts that reduced mean time to resolution by 45%. Participated in architectural design reviews for platform evolution, influencing technical roadmap

decisions that aligned with product goals and customer requirements.

Uber | San Francisco, CA Jun 2021 - Feb 2023

Senior Backend Engineer

Technologies: Go, Java, SQL, MySQL, Kafka, Docker, Kubernetes

- Led development of high-throughput data ingestion platform in Go processing clickstreams and user behavioral events from mobile and web applications, achieving 99.95% uptime and supporting 10 million events per second. Optimized SQL queries and database schema design for analytics workloads, improving query performance by 50% and enabling real-time product analytics dashboards used by 500+ product managers. Mentored team of 5 backend engineers on distributed systems architecture, code review practices, and technical leadership principles, resulting in 3 promotions and improved team retention. Collaborated with frontend teams to design and implement REST APIs for analytics features, ensuring backward compatibility and supporting rapid product iteration cycles.
- Implemented observability infrastructure using distributed tracing and metrics collection, reducing production incident detection time from 30 minutes to 5 minutes. Designed storage solutions for time-series analytics data using specialized databases, reducing storage costs by 35% while maintaining query performance SLAs. Established data quality monitoring systems for clickstream pipelines, detecting and preventing data corruption issues that could impact product analytics accuracy. Leveraged AI coding tools to accelerate development of complex data transformation logic, validating and refining AI-generated code to maintain production quality standards.
- Architected multi-region data replication strategy for analytics platform, ensuring data consistency across geographic regions while maintaining low-latency query performance. Drove technical decisions for query optimization frameworks, implementing cost-based query planning that reduced infrastructure costs by 25%. Participated in on-call rotation and incident response, establishing post-mortem processes that prevented recurring issues and improved system reliability.

Google | Mountain View, CA Aug 2018 - May 2021

Backend Software Engineer

Technologies: Go, Java, SQL, PostgreSQL, Bigtable, Spanner, Protocol Buffers

- Developed backend services in Go and Java for Google Analytics platform, processing clickstreams and user interaction data from billions of websites and mobile applications. Implemented data ingestion pipelines handling 1 million events per second with sub-second latency, utilizing Kafka and custom message queuing systems for reliable delivery. Optimized SQL queries and database schemas for analytics workloads, reducing query latency by 45% and enabling real-time product analytics capabilities for enterprise customers. Collaborated with frontend engineers to design REST APIs and data models supporting analytics visualization and reporting features.
- Designed and implemented observability systems for distributed backend services, including metrics collection, distributed tracing, and log aggregation that improved incident detection and resolution. Mentored junior engineers on backend architecture patterns, code quality standards, and technical problem-solving approaches. Contributed to storage solutions architecture, evaluating and implementing specialized databases for time-series and analytics workloads. Participated in design reviews and technical discussions influencing platform evolution and product roadmap decisions.
- Implemented query optimization techniques including index design, query planning, and caching strategies that improved analytics query performance by 40%. Built data quality monitoring systems detecting anomalies in clickstream data and alerting on data integrity issues. Established best practices for production deployment, monitoring, and incident response across backend services. Leveraged emerging AI tools to improve code review efficiency and identify potential bugs in complex data processing logic.

Amazon Web Services | Seattle, WA Jan 2017 - Jul 2018

Software Engineer II

Technologies: Java, SQL, DynamoDB, S3, Lambda, CloudFormation

- Developed backend services in Java for AWS analytics and monitoring platform, building data ingestion pipelines for customer metrics and observability data. Implemented query optimization strategies for large-scale analytics workloads, improving performance by 35% through intelligent indexing and partitioning. Collaborated with product

teams to design storage solutions supporting diverse analytics use cases, including time-series data, clickstreams, and event logs. Mentored junior engineers on distributed systems design and backend architecture best practices.

- Designed and implemented monitoring and alerting systems for platform reliability, reducing mean time to detection by 40%. Contributed to technical architecture decisions for multi-tenant analytics platform, ensuring data isolation and query performance across customer accounts. Participated in code reviews and design discussions, establishing quality standards and architectural patterns. Built tools and frameworks for data quality validation and anomaly detection in analytics pipelines.
- Optimized storage solutions by implementing tiered storage strategies and compression techniques, reducing infrastructure costs by 30%. Established observability practices including distributed tracing and metrics collection for backend services. Collaborated with frontend teams on API design and data model evolution supporting analytics features.

Microsoft | Redmond, WA Jun 2015 - Dec 2016

Software Engineer

Technologies: Java, C#, SQL Server, Azure, REST APIs

- Developed backend services for Azure analytics platform, implementing data ingestion and processing pipelines for customer telemetry and observability data. Optimized SQL queries and database design for analytics workloads, improving query performance by 30%. Collaborated with cross-functional teams on product analytics features and data-driven decision-making capabilities. Participated in code reviews and technical discussions establishing quality standards.
- Implemented monitoring and alerting systems for platform services, improving reliability and incident detection. Contributed to architecture design for scalable analytics backend supporting millions of concurrent users. Built data quality validation systems for analytics pipelines. Established best practices for production deployment and incident response.
- Designed storage solutions for time-series analytics data, optimizing for query performance and cost efficiency. Mentored junior developers on backend development practices and software engineering principles. Participated in technical design reviews and architecture discussions.

Key Projects

Real-Time Clickstream Analytics Platform

Multi-Tenant Observability Backend

AI-Enhanced Code Review Framework

Certifications

- AWS Certified Solutions Architect Professional
- Google Cloud Professional Data Engineer
- Certified Kubernetes Administrator (CKA)
- Advanced SQL and Query Optimization Specialization
- Distributed Systems Design and Implementation Certificate
- AI Coding Tools and Prompt Engineering Certification
- Datadog Observability Fundamentals Certification

Education

Stanford University

Master of Science 2015 **University of California, Berkeley**

Bachelor of Science 2013